Axiom at AlexandriaX-2026: Confidence-Thresholded Dialectal Encoder Fine-Tuning for MT Error Span Detection and Classification

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Abstract

This paper describes the "Axiom" team's submission to the AlexandriaX-2026 Shared Task, focusing on Subtask 3: MT Error Detection and Classification. We frame the task of predicting fine-grained error spans and their corresponding LQM categories as a joint token classification problem using a BIO tagging scheme. Evaluating multiple Arabic encoders, we find that dialect-specific pretraining significantly outperforms generative extraction baselines. To address severe label imbalance across error categories, we apply a square-root scaled inverse-frequency class-weighted loss. Furthermore, we introduce a confidence-thresholded decoding strategy at inference to reduce false-positive span over-prediction. Our final system achieved highly competitive performance, ranking 5th on the official leaderboard. We highlight that while overlap-based detection is robust, exact span matching remains a persistent challenge due to the non-standardized nature of dialectal Arabic orthography.

1 Introduction and Background

Machine translation (MT) for dialectal Arabic presents unique evaluation challenges. Beyond surface-level fluency, systems must preserve sociolinguistic authenticity and pragmatic intent. The AlexandriaX-2026 Shared Task (El Mekki et al., 2026) introduces a multidimensional evaluation framework to benchmark these capabilities. Subtask 3 requires diagnosing dialectal Arabic MT by identifying exact word-level error spans and assigning an error type based on the Linguistically Motivated Multidimensional Quality Metrics (LQM) typology (Magdy et al., 2026).

While generative information extraction is an active paradigm, exact span localization remains difficult because it requires strict structure, positional fidelity, and boundary control (Xu et al., 2024). Discriminative boundary-aware tagging systems, such as BIO-style sequence labeling, often explicitly encode these constraints more effectively than generative models (Yin and Wang, 2025).

Furthermore, dialectal downstream tasks demand backbones with deep dialectal exposure, as models trained primarily on Modern Standard Arabic (MSA) are often ill-equipped to handle dialectal variation (Abdul-Mageed et al., 2021). Consequently, we approach Subtask 3 as a joint BIO sequence labeling task, evaluating multiple dialect-aware Arabic encoders. We demonstrate that fine-tuning, combined with inference-time confidence thresholding, yields a robust and highly competitive diagnostic system.

2 System Overview

Figure 1 illustrates the overall architecture of our joint detection and classification pipeline. As illustrated, our pipeline processes the machine-translated dialectal Arabic sequence through three primary stages: encoding, projection, and thresholded decoding. First, the input text is subword-tokenized and passed through the `MARBERTv2` encoder to extract contextualized embeddings. These representations are subsequently projected through a linear classification head into a 13-dimensional logits space, representing the probability distribution across all BIO tags. During the training phase, gradients are scaled using our square-root inverse-frequency weighted cross-entropy loss. During inference, a softmax function converts the logits into probabilities, and the confidence filter ($\tau = 0.5$) is applied; any `B-` or `I-` prediction failing to meet this threshold is aggressively reassigned to the `O` (Outside) class. The surviving sequence of tags is then deterministically reconstructed into the final error spans and their corresponding LQM typologies.

Proceedings of The Fourth Arabic Natural Language Processing Conference (ArabicNLP 2026) - Volume 2 Shared Tasks, pages 800–805
October 28-29, 2026 ©2026 Association for Computational Linguistics

2.1 Generative Baseline Attempt

Initially, we explored a generative baseline by applying LoRA fine-tuning to `NileChat-3B-Base` (El Mekki et al., 2025), prompting the model to output JSON arrays containing predicted text spans and categories. This approach underperformed due to schema drift, hallucinated spans, and unreliable boundary mapping when converting generated strings back to original token indices.

2.2 Joint BIO Token Classification

To enforce strict boundary alignment and eliminate generative hallucinations, we reformulated the task as token classification using `MARBERTv2` (Abdul-Mageed et al., 2021) as our backbone encoder, given its strong optimization for dialectal Arabic. Gold character-level spans were mapped to subword token offsets during preprocessing.

The six LQM categories (*graphetics, morphosyntax, orthography_writing_conventions, pragmatics, semantics,* and *sociolinguistics*) were mapped into a 13-class label space: one `O` tag for correct tokens, plus six `B-` tags to mark the beginning of an error span (e.g., `B-semantics`), and six corresponding `I-` tags for span continuation. This allows a single encoder to simultaneously detect the exact error location and classify its type in a single pass.

2.3 Class-Weighted Cross Entropy

The training dataset (1,125 examples) exhibits severe class imbalance, skewed heavily toward *sociolinguistics* and *semantics*. To prevent the model from collapsing into majority classes, we applied an inverse-frequency class-weighted cross-entropy loss. We softened the weights using a square-root scaling factor, which we observed empirically provided a favorable balance between precision and recall compared to standard linear weighting.

2.4 Confidence-Thresholded Decoding

Sequence taggers often suffer from over-prediction on ambiguous text. During inference, we applied a strict confidence threshold. A token is assigned a non-`O` label only if the maximum softmax probability for that class exceeds a predefined threshold ($\tau = 0.5$). This low-cost intervention was observed to reduce false positive span predictions and mitigate boundary bleed during our internal evaluation; any `B-` or `I-` prediction failing to meet this threshold is reassigned to the `O` (Outside) class.

Model Architecture	Dev (Overall)	Test (Overall)
Generative Extraction Baseline		
NileChat-3B-Base (LoRA)	0.205	–
Token Classification Encoders		
AlcLaM-base-v2	0.226	–
CAMeLBERT-mix	0.263	–
CAMeLBERT-da	0.294	0.313
MARBERTv2 (Final)	**0.369**	**0.393**

Table 1: Overall Score comparison across evaluated architectures.

3 Experimental Setup

We evaluated several Arabic encoders to identify the optimal backbone for dialectal span detection.

- **CAMeLBERT-mix** and **CAMeLBERT-da** (Inoue et al., 2021): We evaluate two variants of CAMeLBERT. The *-mix* variant acts as a baseline pretrained on a mixture of MSA, Classical, and dialectal Arabic, while the *-da* variant is pretrained exclusively on dialectal corpora.
- **AlcLaM-base-v2**: An encoder pretrained predominantly on social media data, optimized for dialect tasks (Ahmed et al., 2024).
- **MARBERTv2**: A robust Transformer-base encoder trained on 29B tokens of diverse, highly dialectal Arabic text, previously shown to outperform MSA-centric models on dialectal tasks (Abdul-Mageed et al., 2021).

All models were trained using identical preprocessing pipelines, a fixed random seed of 42, and a maximum sequence length of 256 subword tokens. Detailed training arguments, hyperparameter grids, and computational specifications are provided in Appendix A.

4 Results and Analysis

As summarized in Table 1, our final `MARBERTv2` system achieved an Overall Score of 0.393 on the official blind Test set (with an Overlap F1 of 0.481, Exact Match F1 of 0.166, and Error Class Macro-F1 of 0.304), securing 5th place on the leaderboard.

4.1 Encoder Transferability Insights

Despite `AlcLaM`'s reported strength on sentence-level dialect tasks (e.g., dialect identification) (Ahmed et al., 2024), it underperformed as a token classifier (Dev: 0.226). This supports findings in the literature that social-media pretraining

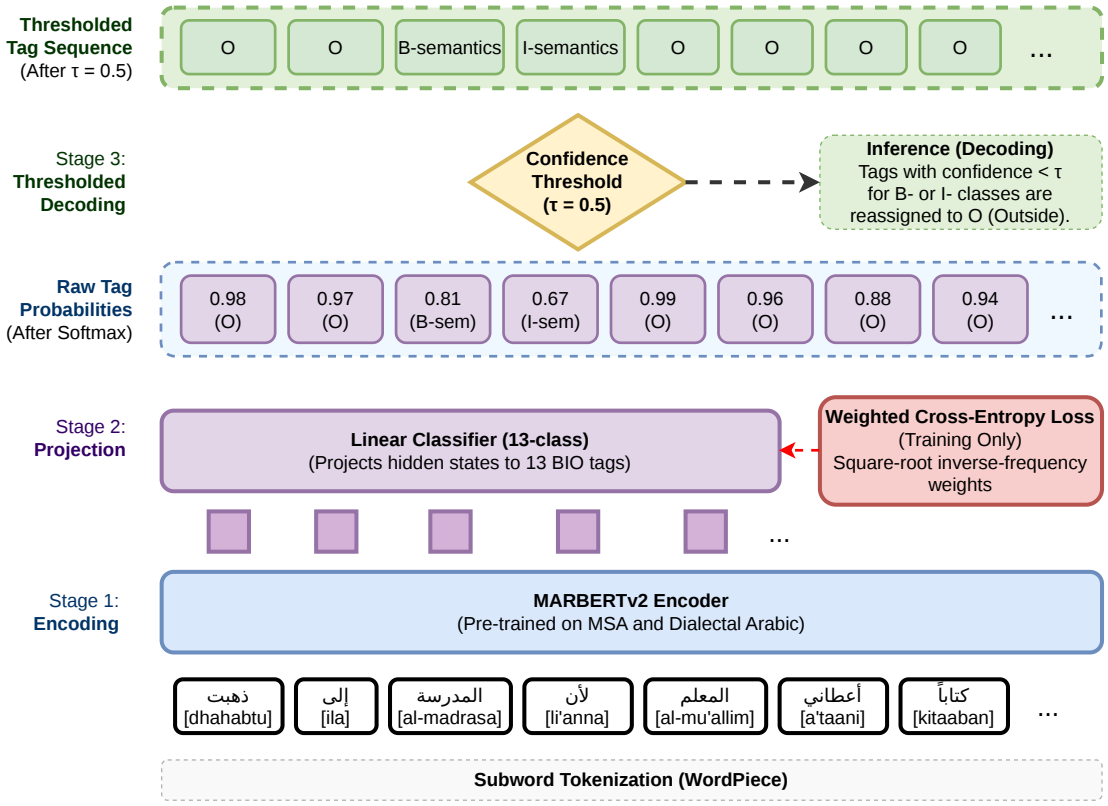


Figure 1: Architecture of our confidence-thresholded BIO token classification pipeline for joint MT error detection and classification.

often helps sentence-level classification (Howard and Ruder, 2018), but does not guarantee successful zero-shot transfer to fine-grained token tagging, where lexical and morphological differences cause sharp performance drops (Khalifa et al., 2021). Conversely, deep dialect-centric encoders (`MARBERTv2` and `CAMeLBERT-da`) substantially outperformed MSA-heavy or general-mix models.

4.2 Error Analysis and The Exact Match Gap

A notable finding is the substantial gap between our system's Overlap F1 (0.481) and Exact Match F1 (0.166). Our model reliably localizes the token region containing an MT error, but pinpointing exact character boundaries remains highly challenging. Dialectal Arabic lacks standardized orthography, and informal grammar, slang, and code-switching make exact span agreement brittle (Ouchene and Bessou, 2026; Alrefaie et al., 2024). This aligns with the task's reported human inter-annotator exact-span agreement (F1 = 0.44), suggesting that exact boundary localization is an inherently noisy sub-problem.

Qualitative analysis of the development set, which yielded 14 strict boundary mismatches indicates that annotation granularity is a highly plausible contributor to this gap, alongside subword-to-character reconstruction artifacts. For instance, human annotators frequently fragment multi-word errors into disjointed spans (occasionally including typographical omissions or whitespace errors), whereas our BIO-tagger extracts cohesive, continuous linguistic clauses. Furthermore, we observed granularity mismatches where annotators flagged entire paragraphs for prolonged register shifts, while our model precisely localized the specific trigger phrases initiating the shift.

Beyond boundary variance, the model occasionally struggles to distinguish structural from stylistic LQM categories. For example, it tends to classify subject-verb agreement mismatches within a phrase context as broad *Sociolinguistics* register shifts, whereas annotators isolate the specific verb as a *Morphosyntax* error. A detailed tabular breakdown of these boundary and category confusion errors, along with English translations, is provided in Appendix B.

5 Conclusion

By framing AlexandriaX-2026 Subtask 3 as a joint BIO token classification problem, we achieved strong localization performance. The combination of dialect-aware encoders (`MARBERTv2`), class-weighted loss, and confidence thresholding proved highly effective, yielding a 5th place ranking. Future work should explore character-level CNNs or CRF layers to improve exact boundary resolution for non-standardized dialects.

Limitations

While our token classification approach demonstrates strong localization, it inherently assumes continuous error spans, which can occasionally diverge from highly fragmented human annotations. Additionally, our confidence threshold ($\tau = 0.5$) was statically optimized for the AlexandriaX-2026 dataset and may require slight recalibration for other dialectal domains. Finally, our sequence length was capped at 256 subword tokens; while highly efficient for standard sentence-level translations, this would need to be extended for document-level applications.

Acknowledgments

The author thanks the organizers of the AlexandriaX-2026 Shared Task for providing the datasets, evaluation framework, and baseline resources that made this research possible.

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A Appendix: Training Details

Models were fine-tuned for a maximum of 20 epochs using the AdamW optimizer with a learning rate of 2e-5 and a weight decay of 0.05. A training batch size of 16 was utilized. Early stopping with a patience of 3 epochs based on validation loss was implemented to mitigate overfitting on the limited 1,125-example training set. The optimal confidence threshold ($\tau = 0.5$) was determined via grid search on the development set probabilities ranging from 0.05 to 0.95.

B Appendix: Error Analysis Examples

Table 2 provides a qualitative breakdown of specific boundary mismatches and category confusions observed in the development set. These examples illustrate how the Exact Match metric heavily penalizes the model for predicting cohesive linguistic units, capturing modifiers, or resolving human annotator fragmentation.

Error Phenomenon	Model Prediction	Gold Reference
Span Fragmentation The model extracted a clean clause. The annotator broke it apart, dropped a final character (ددد), and left a leading space.	نعتقد أنك تريد الدخول (we believe that you want to enter) **[Sociolinguistics]**	Fragmented into 3 spans: 1. نعتقد (we believe) 2. أنك تري (that you wan[t]) 3. الدخول (entering)
Granularity (Verb + Modifier) The model grouped the verb with its direct modifier, while the human isolated only the verb.	فقط قلت (just said) **[Sociolinguistics]**	قلت (said) **[Sociolinguistics]**
Granularity (Trigger vs. Paragraph) The model highlighted the specific phrase triggering an MSA shift, whereas the annotator flagged the entire paragraph block.	وصلتي لأعلى (you reached the highest) **[Sociolinguistics]**	*Entire 40-word paragraph* **[Sociolinguistics]**
Category Ambiguity A subject-verb agreement error. The human flagged it structurally. The model flagged the whole phrase contextually.	ما يُعرف إن كنتُ سنصوم (You do not know if I was fasting [plural]) **[Sociolinguistics]**	كنتُ (I was) **[Morphosyntax]**

Table 2: Representative examples from the Development Set illustrating boundary mismatches and LQM category ambiguity. English translations are provided in parentheses.